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Lab Assignment 3

Q 1: Parsing and Validating Assignment Statements

(Lex file)

```
  GNU nano 9.1                                          Assign.l
%{
#include"y.tab.h"
%}
E               [A-Za-z]
D               [0-9]
ID              (_|{E})(_|{E}|{D})*
NUM             {D}+
%%
{ID}            return(ID);
{NUM}           return(NUM);
[-+*/]  return(*yytext);
[();=\n]                return(*yytext);
%%
```

(Yacc file)

```
  GNU nano 9.1                                  Assign.y
%{
#include<stdio.h>
#include"y.tab.h"
int yylex();
void yyerror(char *s);
%}
%token ID NUM

%%
S       : ID '=' E ';' '\n'             {printf("Given Statement is VALID\n");}
E       : E '+' T
        | E '-' T
        | T
T       : T '*' F
        | T '/' F
        | F
F       : '(' E ')'
        | ID
        | NUM
%%

int main()
{
        printf("Enter the statement: ");
        yyparse();
        return 0;
}

int yywrap()
{
        return 1;
}

void yyerror(char *e)
{
        fprintf(stderr, "Error Message: %s\n", e);
}
```

Output 1:

```
Assign.exe  Assign.l  Assign.y  cdssetup  count.l  lex.yy.c  y.tab.c  y.tab.h
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ yacc -d Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ lex Assign.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ gcc lex.yy.c y.tab.c -o Assign
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ./Assign
Enter the statement: p=q*r;
Given Statement is VALID
```

Output 2:

```
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ yacc -d Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ lex Assign.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ gcc lex.yy.c y.tab.c -o Assign
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ./Assign
Enter the statement: Bhoomi=Compiler+BCI0130*Design;
Given Statement is VALID
```

Q 2: Parsing and Validating Control Flow Statements

(Lex file)

```
  GNU nano 9.1                                          Assign.l
%{
#include"y.tab.h"
%}
E               [A-Za-z]
D               [0-9]
ID              (_|{E})(_|{E}|{D})*
NUM             {D}+
%%
if              return(IF);
while           return(WHILE);
for             return(FOR);
{ID}            return(ID);
{NUM}           return(NUM);
"<"|">"|"<="|">="               return(ROP);
[-+*/]  return(*yytext);
[();={}\n]      return(*yytext);
"++"|"--"               return(IDO);
%%
```

(Yacc file)

```
  GNU nano 9.1                                  Assign.y
%{
#include<stdio.h>
#include"y.tab.h"
int yylex();
void yyerror(char *s);
%}
%token IF WHILE FOR ID NUM IDO ROP

%%
S       : S1 '\n'               {printf("Given IF Statement is VALID\n");}
        | S2 '\n'               {printf("Given WHILE Statement is VALID\n");}
        | S3 '\n'               {printf("Given FOR Statement is VALID\n");}
S1      : IF '(' C ')' '{' ID '=' E ';' '}'
S2      : WHILE '(' C ')' '{' ID '=' E ';' '}'
S3      : FOR '(' ID '=' NUM ';' C ';' ID IDO ')' '{' ID '=' E ';' '}'
C       : IN ROP IN
E       : E '+' T
        | E '-' T
        | T
T       : T '*' F
        | T '/' F
        | F
F       : '(' E ')'
        | IN
IN      : ID
        | NUM
%%

int main()
{
        printf("Enter the statement: ");
        yyparse();
        return 0;
}

int yywrap()
{
        return 1;
}

void yyerror(char *e)
{
        fprintf(stderr, "Error Message: %s\n", e);
}
```

Output 1:

```
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ yacc -d Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ lex Assign.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ gcc lex.yy.c y.tab.c -o Assign
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ./Assign
Enter the statement: for(i=0;BCI0130<=n;i++){Compiler=OS/2;}
Given FOR Statement is VALID
```

Output 2:

```
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ yacc -d Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ lex Assign.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ gcc lex.yy.c y.tab.c -o Assign
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ./Assign
Enter the statement: while(Bhoomi<compiler){design=lex+yacc;}
Given WHILE Statement is VALID
```

Q 3: Simple Calculator using YACC

(Lex file)

```
  GNU nano 9.1                                          Assign.l
%{
#include<stdlib.h>
#include"y.tab.h"
extern int yylval;
%}
D               [0-9]
NUM             {D}+
%%
{NUM}   {yylval=atoi(yytext); return(NUM);}
[-+*/]  return(*yytext);
[()\n]  return(*yytext);
%%
```

(Yacc file)

```
  GNU nano 9.1                                          Assign.y
%{
#include<stdio.h>
#include"y.tab.h"
int yylex();
void yyerror(char *s);
extern int yylval;
%}
%token NUM

%%
S       : E '\n'                {printf("Value: %d",$1);}
E       : E '+' T               {$$ = $1 + $3;}
        | E '-' T               {$$ = $1 - $3;}
        | T                     {$$ = $1;}
T       : T '*' F               {$$ = $1 * $3;}
        | T '/' F               {$$ = $1 / $3;}
        | F                     {$$ = $1;}
F       : '(' E ')'             {$$ = $2;}
        | NUM                   {$$ = $1;}
%%

int main()
{
        printf("Enter the statement: ");
        yyparse();
        return 0;
}

int yywrap()
{
        return 1;
}

void yyerror(char *e)
{
        fprintf(stderr, "Error Message: %s", e);
}
```

Output 1:

```
$ export PS1="Bhoomi Annamdevara 24BCI0130@\h \W \$ "
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ nano count.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ nano Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ls
Assign.exe  Assign.l  Assign.y  cdssetup  count.l  lex.yy.c  y.tab.c  y.tab.h
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ yacc -d Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ lex Assign.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ gcc lex.yy.c y.tab.c -o Assign
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ./Assign
Enter the statement: 01+30
Value: 31
```

Output 2:

```
Error Message: syntax errorBhoomi Annamdevara 24Byacc -d Assign.y
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ lex Assign.l
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ gcc lex.yy.c y.tab.c -o Assign
Bhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ ./Assign
Enter the statement: (01+*30)*24
Error Message: syntax errorBhoomi Annamdevara 24BCI0130@LAPTOP-964OVUQJ ~ $ nano count.l
```